Degradation-Aware Restoration and Quality-Calibrated Object Detection in Hazy and Noisy Environments

Zihao Wang, Xiaolin Zhang*, Ran Liu

Abstract—**Object detection in haze-degraded environments is challenging because haze reduces visibility, weakens target texture and boundary cues, and increases background interference. Consequently, DETR-based detectors may exhibit inadequate representation of degraded inputs, imprecise bounding-box localization, and misalignment between classification confidence and localization quality. To address these issues, we propose DARQ-Det, an end-to-end object detector for haze-degraded scenes. The Degradation-Aware Restoration Module enhances degraded inputs through lightweight residual compensation and total variation regularization, thereby improving the discriminability of target regions. The Hierarchical IoU-Focused Refinement Module introduces IoU-aware focal reweighting to place greater emphasis on low-quality matched boxes during regression optimization. The IoU Quality Calibration Module estimates the localization quality of candidate boxes and combines it with classification confidence to improve candidate ranking. Experiments are conducted on two synthetic haze-degraded remote sensing datasets, RSOD-Haze and HRSC2016-Haze, and the real-world hazy dataset RTTS. DARQ-Det obtains mAP scores of 68.13%, 91.62%, and 60.75% on RSOD-Haze, HRSC2016-Haze, and RTTS, respectively. The results indicate consistent improvements over the RF-DETR baseline in detection accuracy and localization quality across synthetic and real haze-degraded data.**

Index Terms—**Haze-degraded object detection, remote sensing object detection, image restoration, IoU quality calibration, DETR.**

I. Introduction

In recent years, object detection has progressed substantially. Both conventional detection frameworks [1] and end-to-end detectors represented by DETR [2] have achieved strong performance under standard imaging conditions. Their performance may nevertheless deteriorate in remote sensing scenes affected by haze. Targets in remote sensing images are often small, densely distributed, and embedded in complex backgrounds, while haze further reduces contrast, weakens texture and boundary cues, and increases background interference. These factors make feature representation and precise localization more difficult.

Reliable detection in haze-degraded scenes requires both robust classification and accurate localization. This capability is relevant to applications such as remote sensing monitoring, ship detection, and traffic-scene analysis. Sun and Tian [3] reported that traffic-sign detection remains difficult in complex road environments, underscoring the broader need for detectors that remain reliable under degraded visual conditions. Despite recent advances in feature modeling and inference efficiency, haze can still produce three related limitations: inadequate representation of degraded inputs, inaccurate bounding-box localization, and inconsistency between classification confidence and localization quality. RF-DETR [4] offers a suitable baseline because it combines effective backbone representation with a favorable accuracy–efficiency trade-off.

To address these limitations, we propose DARQ-Det, an end-to-end object detection framework that integrates input restoration, localization refinement, and quality calibration. DARQ-Det contains three complementary components. The Degradation-Aware Restoration Module (DARM) applies lightweight residual compensation with total variation regularization to improve the representation of haze-degraded inputs. The Hierarchical IoU-Focused Refinement Module (HIFRM) introduces IoU-aware focal reweighting into bounding-box regression to emphasize low-quality matched boxes. The IoU Quality Calibration Module (IQCM) estimates the localization quality of candidate boxes and uses this estimate to calibrate classification confidence. Together, these components form a progressive pipeline from degraded-input compensation to localization refinement and score calibration.

The main contributions are summarized as follows:

(1) We propose DARQ-Det, an end-to-end object detection framework for haze-degraded scenes that jointly addresses degraded-input representation, bounding-box localization, and prediction-score calibration.

(2) We develop DARM, which combines lightweight residual compensation with total variation regularization to improve the representation of haze-degraded images.

(3) We introduce HIFRM, which incorporates IoU-aware focal reweighting into bounding-box regression to place greater emphasis on low-quality matched boxes.

(4) We design IQCM to estimate localization quality and combine it with classification confidence, thereby improving candidate ranking and the reliability of the final detections.

II. Related Work

A. Non-DETR Object Detection and Backbone Feature Representation

Object detection methods can be broadly grouped into conventional non-DETR approaches [1] and DETR-based approaches [2]. Non-DETR methods include two-stage detectors [1] and one-stage detectors [5], with subsequent

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research improving their accuracy, feature representation, and inference efficiency. Hou et al. [6] investigated lightweight object detection for remote sensing images, while Zhang et al. [7] developed POD-YOLO with bidirectional dynamic cross-level feature aggregation. These studies indicate that efficient and discriminative feature representation remains important, particularly in complex scenes and resource-constrained settings.

Faster R-CNN is a representative two-stage detector that incorporates a Region Proposal Network into the detection pipeline. The proposal and detection stages share convolutional features, allowing region generation, object classification, and bounding-box regression to be optimized within a unified framework. In parallel, advances in visual backbones have further improved the quality of representations used by detection models.

Swin Transformer [8] constructs hierarchical feature representations and uses shifted-window self-attention to exchange information across local windows while controlling computational cost. This design makes Transformer backbones suitable for dense prediction tasks such as object detection. However, many conventional detection frameworks continue to rely on anchor design and post-processing, which increases pipeline complexity and motivates the development of end-to-end alternatives.

B. DETR-Based End-to-End Detection and Real-Time Advances

DETR [2] formulates object detection as a set-prediction problem and uses bipartite matching to produce detections without anchor design or non-maximum suppression. Subsequent studies have focused primarily on training convergence, query initialization, detection accuracy, and real-time inference.

DINO [9] improves DETR training and detection performance through contrastive denoising, mixed query selection, and a two-stage look-forward mechanism. Its design primarily emphasizes detection accuracy, with less emphasis on real-time deployment.

RT-DETR [10] addresses real-time end-to-end detection through an efficient hybrid encoder and an improved query-initialization strategy. RT-DETRv2 [11] further improves training flexibility and deployment practicality. Nevertheless, efficient representation modeling and lightweight architectural design remain relevant concerns for real-time DETR variants.

LW-DETR [12] adopts a compact encoder–decoder architecture to balance detection accuracy and real-time performance. Building on this line of work, RF-DETR [4] combines a DINOv2 backbone with weight-sharing neural architecture search to improve feature representation and the accuracy–latency trade-off. These properties make RF-DETR an appropriate baseline for studying input restoration, localization refinement, and quality calibration in haze-degraded scenes.

C. Object Detection under Haze and Other Visual Degradations

Visual degradations such as haze and low illumination reduce visibility and obscure image details, thereby increasing the difficulty of object detection. Earlier studies commonly applied image enhancement or dehazing as a preprocessing step. Because restoration and detection are optimized separately in such pipelines, the restored images are not necessarily aligned with the detector's objectives.

IA-YOLO [13] integrates image enhancement with object detection and optimizes both components jointly. The method is primarily designed for natural scenes and does not specifically address the characteristics of haze-degraded remote sensing imagery. For remote sensing applications, DHC-Net [14] combines adaptive dehazing with contrastive learning to mitigate image degradation and class imbalance.

Existing work has mainly examined front-end restoration, feature enhancement, or their integration with conventional detectors. Comparatively less attention has been given to adapting end-to-end DETR frameworks to haze-degraded scenes. We therefore use RF-DETR as the baseline and incorporate input restoration, localization refinement, and quality calibration within a unified end-to-end framework.

D. Localization Optimization and Quality Calibration

Beyond degraded feature representation, detection performance in hazy scenes is also affected by imprecise bounding-box localization and misalignment between classification confidence and localization quality. DETR-based methods commonly use L_1 and GIoU losses [15] for box regression. Although this formulation is broadly applicable, it does not explicitly differentiate matched boxes according to localization difficulty. Under haze, blurred contours and increased background interference can make ambiguous matches more difficult to refine.

Previous studies have incorporated IoU information into classification objectives or ranking strategies to reduce classification–localization misalignment. For example, LW-DETR [12] uses an IoU-aware classification loss so that the classification objective reflects localization quality during training. This approach suggests that candidate ranking based solely on classification confidence may be suboptimal and that explicit quality information can improve ranking reliability.

Although these approaches improve box regression or classification–localization consistency, most are developed for general object detection and do not explicitly consider haze-induced degradation. Accordingly, we introduce targeted modifications to RF-DETR for localization refinement and quality calibration in haze-degraded scenes.

III. Methodology

A. Overall Network Architecture

DARQ-Det extends the RF-DETR baseline by introducing three components that address degraded-input representation, bounding-box localization, and prediction-score calibration. The framework combines input restoration, localization refinement, and quality calibration within a single end-to-end detection pipeline.

As shown in Fig. 1, the framework comprises DARM, HIFRM, and IQCM. The input image I is first processed by DARM to obtain the restored image I'. The backbone then extracts multi-level features $\{F_l\}$, where C_b denotes the channel dimension of the backbone output. The Projector aligns the channels, adapts the feature scales, and fuses the

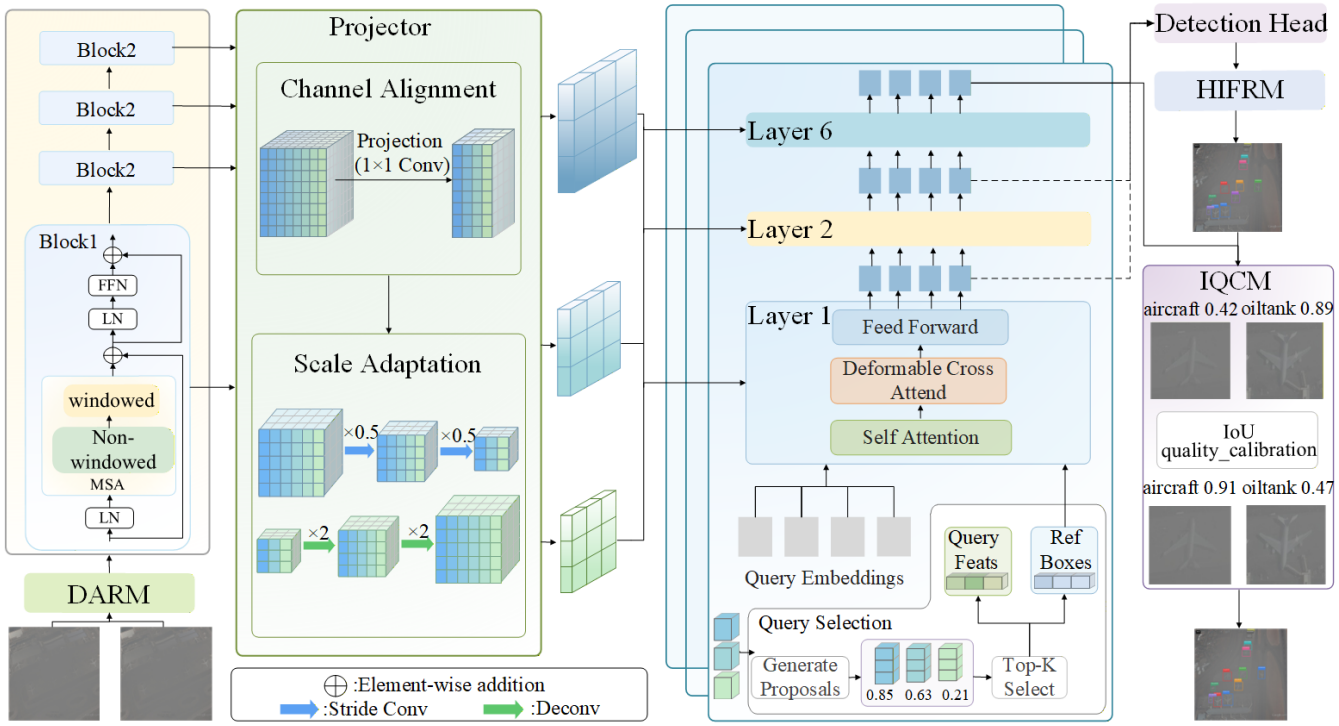


Fig. 1. Overall architecture of DARQ-Det, comprising DARM, HIFRM, and IQCM.

projected features at the target resolution to generate the decoder input F_d, whose channel dimension is denoted by C_d. The decoder jointly models F_d and N_q object queries to produce query features Q. The detection head subsequently generates classification predictions P_{cls} and bounding-box predictions B_{pred} for C target categories.

The training and inference paths differ after decoding. During training, HIFRM uses B_{pred} and the ground-truth boxes to compute the localization objective, while IQCM learns to estimate the localization quality of selected candidate boxes from the query features Q. During inference, IQCM predicts localization-quality scores S_{IoU} and combines them with P_{cls} to obtain the calibrated scores P_{final}. The final detections are produced from P_{final} and B_{pred}. The following subsections describe the three modules in detail.

B. Degradation-Aware Restoration Module

Haze reduces target-region contrast and weakens the visual cues available to the detector. DARM is therefore designed to compensate for degradation at the input stage before feature extraction. Its structure is shown in Fig. 2.

Let the input image be $I \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the image height, width, and number of channels, respectively. Three convolutional layers and two GELU activations map I to a restoration residual $R \in \mathbb{R}^{H\times W\times C}$:

$$R = \text{Conv}_3\big(\text{GELU}(\text{Conv}_2(\text{GELU}(\text{Conv}_1(I))))\big), \quad (1)$$

where $\text{Conv}_i(\cdot)$ denotes the i-th 3×3 convolution and $\text{GELU}(\cdot)$ is the GELU activation function. The residual map R encodes local corrections used in the subsequent residual fusion.

The residual map is scaled by a coefficient α and added element-wise to the input image, yielding the restored image $I' \in \mathbb{R}^{H\times W\times C}$:

$$I' = I + \alpha R, \quad (2)$$

where α is the residual-adjustment coefficient that controls the strength of the compensation. The residual connection limits the modification to an additive correction and helps preserve the global structure of the input image.

During training, a total variation term is applied to R to encourage spatial smoothness:

$$L_{\text{TV}} = \sum_{i,j} \left(|R_{i+1,j} - R_{i,j}| + |R_{i,j+1} - R_{i,j}|\right), \quad (3)$$

where L_{TV} denotes the total variation regularization term and (i, j) indexes a spatial location in the residual map. The term penalizes abrupt horizontal and vertical changes in R. It is used only during training; inference retains only residual generation and fusion. In this way, DARM provides a constrained input-level correction before backbone feature extraction.

C. Hierarchical IoU-Focused Refinement Module

Haze can obscure target boundaries and increase localization uncertainty. HIFRM is designed to place greater regression emphasis on matched boxes with poor overlap quality. Its structure is illustrated in Fig. 3.

The predicted and ground-truth boxes are first associated through Hungarian matching. For the i-th predicted box $\hat{b}_i$ and the j-th ground-truth box b_j, the matching cost is defined as

$$\mathcal{C}_{i,j} = \alpha_{\text{cls}}\mathcal{C}^{\text{cls}}_{i,j} + \alpha_{\text{L1}}\|\hat{b}_i - b_j\|_1 + \alpha_{\text{GIoU}}(1 - \text{GIoU}(\hat{b}_i, b_j)), \quad (4)$$

where $\mathcal{C}^{\text{cls}}_{i,j}$ is the classification cost, $\left\|\hat{b}_i - b_j\right\|_1$ is the L_1 distance, and $\text{GIoU}(\hat{b}_i, b_j)$ is the generalized intersection over

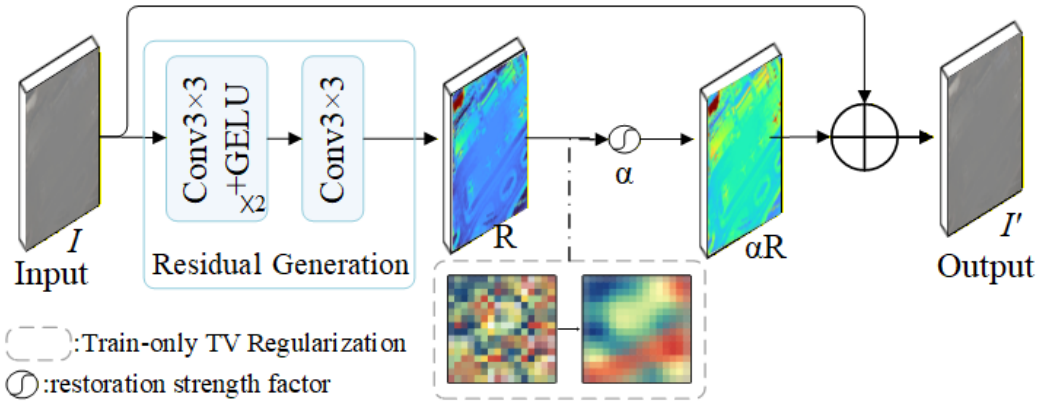


Fig. 2. Structure of the Degradation-Aware Restoration Module (DARM). DARM predicts a restoration residual through convolutional mapping, scales it using a restoration-strength coefficient, and adds it to the input image. Total variation regularization is applied only during training.

union. The coefficients α_{cls}, α_{L1}, and α_{GIoU} weight the three terms. Following Deformable-DETR [16] and DN-DETR [17], they are set to 2, 5, and 2, respectively.

The optimal one-to-one assignment is obtained by minimizing the total matching cost:

$$\hat{\sigma} = \arg\min_{\sigma} \sum_{j=1}^{N_{\text{gt}}} \mathcal{C}_{\sigma(j),j}, \tag{5}$$

where N_{gt} is the number of ground-truth objects, $\sigma(j)$ is the index of the prediction assigned to the j-th ground-truth box, and $\hat{\sigma}$ is the optimal assignment. The resulting matched pairs are used for regression supervision.

For each matched pair, HIFRM applies focal reweighting to the GIoU loss so that the contribution of a sample depends on its overlap quality. The original GIoU loss for the k-th pair is

$$\mathcal{L}_{\text{GIoU}}^{(k)} = 1 - \text{GIoU}(\hat{b}_k, b_k), \tag{6}$$

where $\hat{b}_k$ and b_k denote the predicted and ground-truth boxes in the k-th matched pair, respectively. The focal-weighted term is

$$\tilde{\mathcal{L}}_{\text{GIoU}}^{(k)} = \left(\mathcal{L}_{\text{GIoU}}^{(k)}\right)^{\gamma}, \tag{7}$$

where γ is the focal modulation coefficient. It controls how strongly the loss differentiates samples according to localization quality. For $\gamma > 1$, larger GIoU losses receive greater relative emphasis, directing optimization toward harder-to-localize matched boxes.

The regression terms used by HIFRM are then defined as

$$\mathcal{L}_{\text{L1}} = \sum_{k=1}^{N_{\text{gt}}} \left\| \hat{b}_k - b_k \right\|_1, \tag{8}$$

$$\tilde{\mathcal{L}}_{\text{GIoU}} = \sum_{k=1}^{N_{\text{gt}}} \tilde{\mathcal{L}}_{\text{GIoU}}^{(k)}, \tag{9}$$

$$\mathcal{L}_{\text{reg}} = \beta_{\text{L1}}\mathcal{L}_{\text{L1}} + \beta_{\text{GIoU}}\tilde{\mathcal{L}}_{\text{GIoU}}, \tag{10}$$

where $\mathcal{L}_{\text{L1}}$ constrains box position and scale, $\tilde{\mathcal{L}}_{\text{GIoU}}$ is the focal-weighted GIoU term, and β_{L1} and β_{GIoU} are their respective weights. Following DN-DETR [17], these coefficients are set to 5 and 2. HIFRM is applied to the box predictions from all decoder layers. This multi-stage supervision is the hierarchical component of the module and progressively constrains localization throughout decoding.

D. IoU Quality Calibration Module

In haze-degraded scenes, a candidate box may receive a high classification score even when its localization quality is limited. Such misalignment can adversely affect candidate ranking. IQCM therefore estimates localization quality explicitly and uses the estimate to calibrate classification confidence.

During the decoding stage, let the query features output by the last decoder layer be denoted as

$$H = \{h_i\}_{i=1}^{N_q}, \quad h_i \in \mathbb{R}^{C_q}, \tag{11}$$

where N_q denotes the number of queries and C_q is the feature dimension of each query. A lightweight quality head maps each final-layer query feature to an IoU-quality score:

$$\hat{q}_i = \sigma(W_q h_i + b_q), \tag{12}$$

where W_q and b_q are the weight and bias of the linear mapping, $\sigma(\cdot)$ is the sigmoid function, and $\hat{q}_i \in [0, 1]$ is the predicted localization-quality score for the i-th query.

To limit supervision from low-confidence background queries, IQCM selects only the top M queries according to their classification responses. Let $z_i \in \mathbb{R}^C$ be the classification logits of the i-th query, where C is the number of target categories. Its objectness score is defined as

$$s_i = \max_{c \in \{1,\ldots,C\}} \sigma(z_{i,c}), \tag{13}$$

where s_i is the maximum classification response across the target categories. The indices of the top M queries form the quality-supervision set:

$$\mathcal{I}_{\text{top}} = \text{TopM}\left(\{s_i\}_{i=1}^{N_q}, M\right). \tag{14}$$

For each selected query $i \in \mathcal{I}_{\text{top}}$, let $\hat{b}_i$ denote its predicted box and define the ground-truth box set as

$$B = \{b_j\}_{j=1}^{N_t}, \tag{15}$$

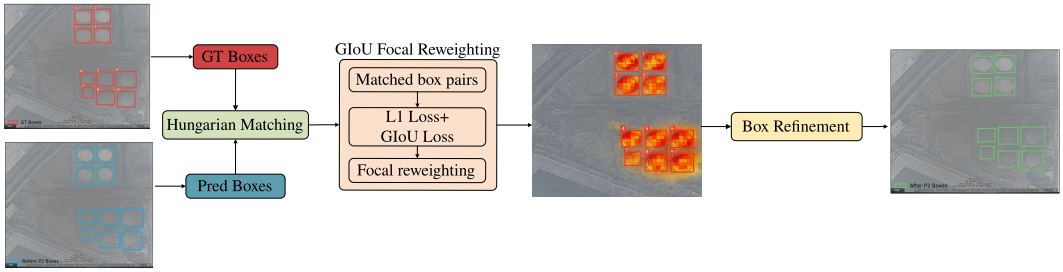


Fig. 3. Structure of the Hierarchical IoU-Focused Refinement Module (HIFRM). HIFRM establishes one-to-one box correspondences through Hungarian matching and applies GIoU-based focal reweighting to the matched pairs during regression optimization.

where N_t is the number of ground-truth objects in the image. The maximum IoU between $\hat{b}_i$ and the ground-truth boxes is used as the quality target:

$$q_i^* = \begin{cases} \max\limits_{1\le j\le N_t} \mathrm{IoU}(\hat{b}_i, b_j), & N_t > 0, \\ 0, & N_t = 0, \end{cases} \tag{16}$$

where $q_i^* \in [0,1]$ is the localization-quality target for the i-th query. This target allows the quality branch to learn box overlap directly rather than infer it only from classification confidence.

IQCM uses binary cross-entropy with the IoU targets to supervise the predicted quality scores:

$$\mathcal{L}_{\mathrm{IQC}} = -\frac{1}{M} \sum_{i\in\mathcal{I}_{\mathrm{top}}} \left[q_i^* \log(\hat{q}_i) + (1-q_i^*) \log(1-\hat{q}_i) \right], \tag{17}$$

where $\mathcal{L}_{\mathrm{IQC}}$ denotes the IoU-quality calibration loss. This objective encourages the predicted quality score to reflect the overlap of the corresponding box.

Finally, the IQCM loss term is defined as

$$\mathcal{L}_{\mathrm{IQCM}} = \lambda_{\mathrm{IQCM}} \mathcal{L}_{\mathrm{IQC}}, \tag{18}$$

where λ_{IQCM} is the weight assigned to the quality-calibration loss. The auxiliary objective adds localization-quality supervision without altering the main decoder structure. During inference, the predicted quality scores are combined with the classification confidence so that candidate ranking better reflects the estimated localization quality of each box.

IV. Experimental Design and Implementation

A. Dataset and Haze Construction

DARQ-Det is evaluated on three datasets: the synthetic haze-degraded remote sensing datasets RSOD-Haze and HRSC2016-Haze, and the real-world hazy dataset RTTS. The first two datasets are generated offline from RSOD and HRSC2016, respectively, while RTTS is used to assess performance under naturally occurring haze.

RSOD [18]. RSOD is a public remote sensing object detection dataset containing 976 images and 6,950 annotated instances from four categories. It includes multi-scale targets and complex backgrounds and is therefore commonly used to evaluate remote sensing detectors.

HRSC2016 [19]. HRSC2016 is a ship-detection benchmark containing 1,061 images with resolutions from 300×300 to 1500×900 pixels. In our experiments, 436 images are used for training and 181 for testing. All images are resized to 800×800 pixels before being fed into the model.

RTTS [20]. RTTS is a publicly available dataset for object detection in real-world hazy scenes and is part of the RESIDE-β benchmark. It contains 4,322 traffic-related images annotated in a VOC 2007-compatible format for five categories: person, bicycle, car, bus, and motorbike. Because no paired haze-free reference images are provided, RTTS is used here to evaluate detection performance under real haze rather than restoration fidelity.

B. Synthetic Haze Data Generation Strategy

Because the original RSOD and HRSC2016 datasets contain clear remote sensing images, we apply offline haze synthesis to construct RSOD-Haze and HRSC2016-Haze. The synthesis procedure follows the atmospheric scattering model used in IA-YOLO [13] and DHC-Net [14].

Let $J(x)$ and $I(x)$ denote the clear image and the hazy image at pixel location x, respectively. The hazy imaging process is formulated as

$$I(x) = J(x)t(x) + A\big(1 - t(x)\big), \tag{19}$$

where A denotes the global atmospheric light and $t(x)$ denotes the transmission map. The transmission map is defined as

$$t(x) = e^{-\beta d(x)}, \tag{20}$$

where β is the atmospheric scattering coefficient and $d(x)$ is the scene depth. This formulation models the attenuation of scene radiance and the contribution of atmospheric light and is commonly used for synthetic haze generation.

Because RSOD and HRSC2016 do not provide depth maps, a radial approximation is used to obtain a spatially varying depth field:

$$d(x) = 0.04 \frac{\rho(x)}{\max(H, W)}, \tag{21}$$

where $\rho(x)$ is the Euclidean distance from pixel x to the image center, and H and W are the image height and width. The resulting transmission map varies spatially without requiring depth annotations, producing nonuniform visibility attenuation across the image.

In the implementation, the atmospheric light is fixed at $A = 0.05$. Light, moderate, and heavy haze are generated using $\beta \in [0.003, 0.006]$, $[0.013, 0.015]$, and $[0.018, 0.020]$, respectively. The same offline synthesis procedure is applied to the training, validation, and test partitions. Because the transformation changes image appearance but not object

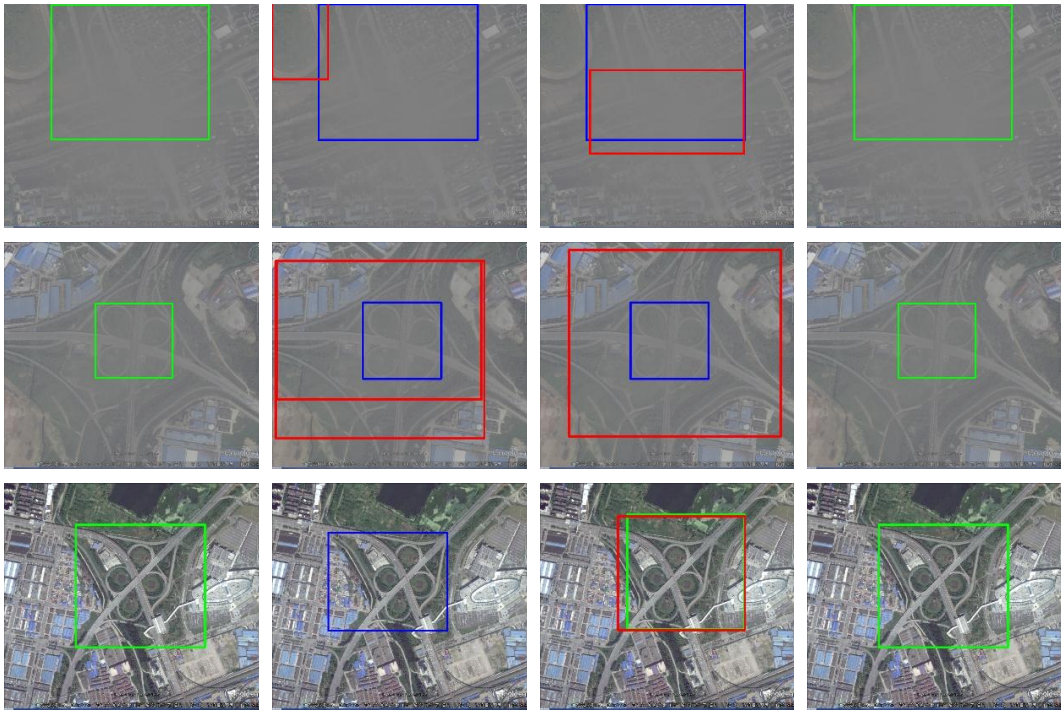


Fig. 4. Qualitative results on representative RSOD-Haze test images: (a) ground truth, (b) RT-DETR, (c) RF-DETR, and (d) DARQ-Det.

categories or locations, the original bounding-box annotations are retained for the corresponding hazy images.

To evaluate the effect of haze-density diversity, the adopted multi-level strategy is compared with a single-level setting in which β is fixed at 0.014, while all other experimental settings remain unchanged.

TABLE I
ABLATION STUDY OF THE MULTI-LEVEL HAZE SYNTHESIS STRATEGY ON THE RSOD-HAZE VALIDATION SET.

Haze synthesis strategy	β setting	mAP	AP_{50}	AP_{75}
Single-level haze	0.014	66.57	95.02	74.98
Multi-level haze	Three predefined ranges	68.13	96.63	76.61

As shown in Table I, the multi-level haze synthesis strategy improves mAP, AP_{50}, and AP_{75} by 1.56, 1.61, and 1.63 percentage points, respectively, compared with the single-level setting. These consistent gains suggest that exposure to light, moderate, and heavy haze during training improves the detector's adaptability to variations in visibility, contrast, and boundary degradation. In contrast, the fixed β setting represents only a single degradation pattern, which may limit generalization across different haze densities.

C. Evaluation Metrics

We report mean average precision (mAP), AP_{50}, and AP_{75}. At a fixed IoU threshold, average precision (AP) is the area under the precision–recall curve and reflects classification, localization, and confidence ranking jointly. Reporting AP at multiple IoU thresholds is useful for remote sensing detection, where small and densely distributed objects can make localization errors particularly consequential.

The maximum number of detections per image is set to 1500 to accommodate densely distributed objects. At a specified IoU threshold, AP is defined as

$$\mathrm{AP} = \int_0^1 P(r)\, dr, \tag{22}$$

where r denotes recall, and $P(r)$ represents the precision at a given recall level. Recall and precision are computed as follows:

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{23}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \tag{24}$$

where true positives (TP) are predictions matched to ground-truth objects at the specified Intersection over Union (IoU) threshold, false positives (FP) are unmatched predictions, and false negatives (FN) are ground-truth objects without a matched prediction.

For evaluation, predictions are sorted by confidence and matched one-to-one with previously unmatched ground-truth boxes of the same category. A prediction is counted as a TP when its maximum IoU satisfies the selected threshold; otherwise, it is counted as an FP. Ground-truth boxes that remain unmatched are counted as FNs. This procedure evaluates classification and localization jointly.

The metric AP_{50} uses an IoU threshold of 0.50 and primarily reflects object discovery under a moderate localization requirement. The metric AP_{75} uses a threshold of 0.75 and is more sensitive to bounding-box accuracy. The reported mAP is the mean of AP values computed at IoU thresholds from 0.50 to 0.95 in increments of 0.05. Together, these metrics characterize overall detection accuracy and localization quality under haze degradation.

D. Implementation Details

All experiments are conducted on a server equipped with an NVIDIA RTX 3090 GPU. For controlled comparisons, the evaluated variants follow the architecture and training configuration of RF-DETR Medium. The baseline uses a windowed DINOv2 visual encoder for multi-scale feature extraction. The hidden dimension of the detection branch is 256, and the feed-forward network dimension is 2048.

The models are trained for 180 epochs using AdamW [21]. The learning rates for the main network and encoder are set to 1×10^{-4} and 1.5×10^{-4}, respectively, with a weight decay of 1×10^{-4}. The batch size is 4, and gradients are accumulated over four steps. A StepLR scheduler reduces the learning rate after epoch 100. Unless stated otherwise, the remaining settings are identical to those of the RF-DETR baseline.

E. Main Results

Comparative experiments are conducted on RSOD-Haze, HRSC2016-Haze, and RTTS using representative non-DETR and DETR-based detectors. RSOD-Haze and HRSC2016-Haze provide controlled synthetic degradation, whereas RTTS evaluates the method under real haze. Across the three datasets, DARQ-Det consistently improves mAP over the RF-DETR baseline, although the magnitude of the gain varies by dataset and metric.

TABLE II
EXPERIMENTAL RESULTS ON THE RSOD-HAZE DATASET. ALL MODELS ARE TRAINED ON THE TRAINING SET AND EVALUATED ON THE TEST SET. * DENOTES RE-IMPLEMENTED RESULTS.

Method	Parameters (M)	mAP	AP_{50}	AP_{75}
Non-DETR methods				
Faster R-CNN [1] *	37.61	45.4	82.7	44.7
Swin-T [8] *	45.32	55.8	92.1	62.4
LSKNet-T [22] *	39.33	57.2	93.3	63.7
DETR-based methods				
OWRT-DETR [23] *	38.01	50.52	81.26	55.32
UAV-DETR [24] *	44.62	58.30	85.56	64.36
LW-DETR [12] *	28.23	58.8	88.0	60.7
DINO-DETR [9] *	46.67	60.1	87.5	69.1
RT-DETR [10] *	42.71	62.6	92.4	71.2
RT-DETRv2 [11] *	36.41	64.4	93.6	69.7
RF-DETR [4] *	33.36	67.02	95.51	75.13
DARQ-Det (Ours)	34.44	68.13	96.63	76.61

RSOD-Haze. Table II reports the results on the multi-category RSOD-Haze dataset. DARQ-Det obtains the highest mAP, AP_{50}, and AP_{75} among the compared methods, with values of 68.13%, 96.63%, and 76.61%, respectively.

Relative to the non-DETR methods, DARQ-Det yields higher mAP under the same haze-degraded evaluation setting. This result indicates that the proposed framework remains effective when target contrast and boundary information are reduced. Because the compared methods differ in architecture and training design, the comparison should be interpreted as an overall performance comparison rather than as evidence for a single causal factor.

Compared with RF-DETR, DARQ-Det improves mAP, AP_{50}, and AP_{75} by 1.11, 1.12, and 1.48 percentage points, respectively. The larger gain at AP_{75} is consistent with an improvement in stricter localization performance. Relative to RT-DETRv2, the corresponding gains are 3.73, 3.03, and 6.91 percentage points. These improvements are obtained with 34.44 M parameters, only 1.08 M more than RF-DETR, indicating that the performance gain is achieved with a limited increase in model size.

TABLE III
EXPERIMENTAL RESULTS ON THE HRSC2016-HAZE DATASET. ALL MODELS ARE TRAINED ON THE TRAINING SET AND EVALUATED ON THE TEST SET. * DENOTES RE-IMPLEMENTED RESULTS.

Method	Parameters (M)	mAP	AP_{50}	AP_{75}
Non-DETR methods				
Faster R-CNN [1] *	37.61	67.0	93.2	78.9
LSKNet-T [22] *	39.33	76.3	94.3	88.8
Swin-T [8] *	45.32	78.4	93.3	89.9
DETR-based methods				
OWRT-DETR [23] *	38.01	84.99	94.65	90.88
UAV-DETR [24] *	44.62	87.12	95.17	91.96
LW-DETR [12] *	28.23	87.9	95.9	93.0
DINO-DETR [9] *	46.67	89.4	96.9	93.8
RF-DETR [4] *	33.36	90.46	97.53	95.5
RT-DETR [10] *	42.71	90.9	97.6	95.3
RT-DETRv2 [11] *	36.41	91.0	97.3	95.3
DARQ-Det (Ours)	34.44	91.62	98.22	96.43

HRSC2016-Haze. Table III presents the ship-detection results on HRSC2016-Haze. The elongated geometry and boundary sensitivity of ship targets make this dataset useful for assessing localization under degraded visibility. DARQ-Det records the highest mAP, AP_{50}, and AP_{75} among the compared methods, reaching 91.62%, 98.22%, and 96.43%, respectively. It also uses fewer parameters than RT-DETR and RT-DETRv2, suggesting that the improvement is not obtained solely through a larger model.

Compared with Faster R-CNN and Swin Transformer, DARQ-Det improves mAP by 24.62 and 13.22 percentage points, respectively. The corresponding AP_{75} gains are 17.53 and 6.53 percentage points, indicating a substantial difference under the stricter localization criterion. These results are consistent with the intended role of degradation-aware compensation and localization-focused regression, although architectural differences among the compared methods also contribute to the observed gaps.

Among the DETR-based methods, DARQ-Det improves mAP, AP_{50}, and AP_{75} over RF-DETR by 1.16, 0.69, and 0.93 percentage points, respectively. Relative to RT-DETRv2, the gains are 0.62, 0.92, and 1.13 percentage points. Because most DETR-based methods already achieve high AP_{50} values on this dataset, the improvement in AP_{75} provides additional evidence of better localization under a stricter overlap requirement.

The pattern of improvements is consistent with the complementary design of the three modules: DARM modifies the degraded input before feature extraction, HIFRM reweights box-regression errors, and IQCM incorporates

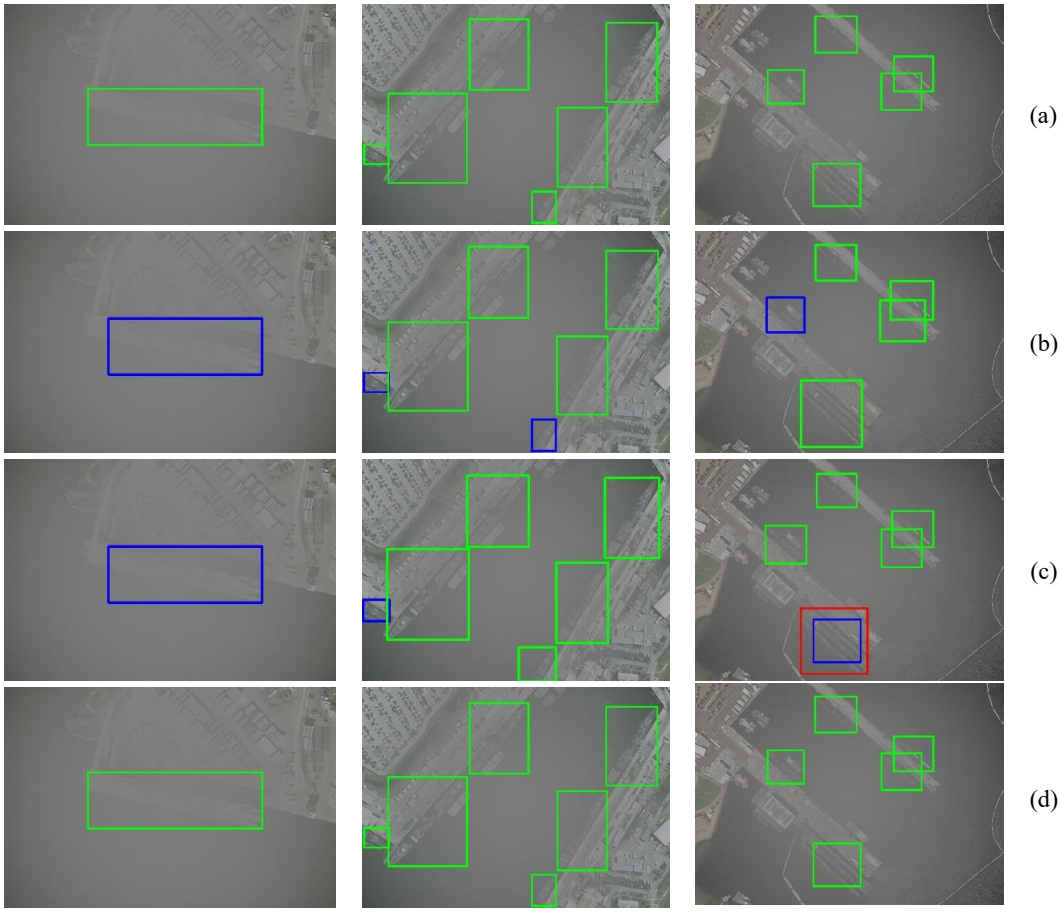


Fig. 5. Qualitative results on representative HRSC2016-Haze test images: (a) ground truth, (b) RT-DETR, (c) RF-DETR, and (d) DARQ-Det.

estimated localization quality into candidate ranking.

RTTS. Table IV reports results on real hazy traffic scenes. DARQ-Det obtains 60.75% mAP, 85.06% AP_{50}, and 68.10% AP_{75}. It achieves the highest mAP and AP_{50} among the compared methods, while RT-DETRv2 is 0.10 percentage points higher in AP_{75}.

TABLE IV
EXPERIMENTAL RESULTS ON THE RTTS DATASET. ALL MODELS ARE TRAINED ON THE TRAINING SET AND EVALUATED ON THE TEST SET.

Method	Parameters (M)	mAP	AP_{50}	AP_{75}
Non-DETR methods				
Faster R-CNN [1]	37.61	44.7	77.4	45.8
Swin-T [8]	45.32	45.9	72.3	52.0
LSKNet-T [22]	39.33	46.9	73.6	52.2
DETR-based methods				
RT-DETR [10]	42.71	59.8	83.7	67.1
RT-DETRv2 [11]	36.41	60.2	84.3	68.2
OWRT-DETR [23]	38.01	49.32	73.32	54.77
UAV-DETR [24]	44.62	54.06	77.33	60.42
DINO-DETR [9]	46.67	55.1	78.9	63.3
LW-DETR [12]	28.23	59.0	83.7	65.3
RF-DETR [4]	33.36	59.9	84.6	67.6
DARQ-Det (Ours)	34.44	60.75	85.06	68.10

Compared with Faster R-CNN and LSKNet-T, DARQ-Det improves mAP by 16.05 and 13.85 percentage points, respectively. The gains are also observed in AP_{50} and AP_{75}. These results show that the proposed framework remains competitive in scenes with naturally varying haze and complex road backgrounds.

Compared with RT-DETRv2, DARQ-Det improves mAP and AP_{50} by 0.55 and 0.76 percentage points, respectively, but records a slightly lower AP_{75}. Relative to RF-DETR, the gains in mAP, AP_{50}, and AP_{75} are 0.85, 0.46, and 0.50 percentage points. The consistent gains over the baseline suggest that the proposed modifications are not limited to the synthetic haze-generation procedure.

The RTTS results are consistent with the intended functions of DARM, HIFRM, and IQCM, which respectively address degraded-input compensation, box-regression weighting, and quality-aware candidate ranking. Because the gains on RTTS are smaller than those on the synthetic datasets, they should be interpreted as moderate but consistent improvements in real haze. The smaller margin may be related to the broader variation in natural haze, illumination, and traffic backgrounds, which is not fully represented by a controlled synthesis process. Moreover, the slight AP_{75} disadvantage relative to RT-DETRv2 indicates that the benefit is not uniform at every localization threshold. Overall, the results indicate consistent performance gains across synthetic and real haze conditions and motivate the following component-wise analysis.

F. Ablation Studies

DARQ-Det contains DARM, HIFRM, and IQCM, which target degraded-input representation, bounding-box localization, and classification–localization consistency, respectively. Ablation studies are conducted on RSOD-Haze, HRSC2016-Haze, and RTTS using RF-DETR as the baseline. The three datasets cover synthetic remote sensing haze and real-world traffic haze.

RSOD-Haze. Table V shows that each module improves mAP when added individually to RF-DETR. DARM increases mAP from 67.02% to 67.21%, HIFRM to 67.18%, and IQCM to 67.13%. These modest but consistent changes are aligned with the respective objectives of input compensation, localization-focused regression, and quality-aware calibration.

TABLE V
OVERALL ABLATION RESULTS OF DARQ-DET ON THE RSOD-HAZE VALIDATION SET.

DARM	HIFRM	IQCM	Params (M)	mAP	AP_{50}	AP_{75}
–	–	–	33.36	67.02	95.51	75.13
✓	–	–	33.41	67.21	95.73	75.70
–	✓	–	33.38	67.18	95.66	75.42
–	–	✓	33.41	67.13	95.60	75.38
✓	✓	–	33.67	67.75	95.77	76.64
–	✓	✓	33.71	67.71	95.76	76.45
✓	–	✓	33.88	67.66	95.75	76.23
✓	✓	✓	34.44	68.13	96.63	76.61

Two-module combinations yield larger mAP gains than the corresponding single-module variants. DARM+HIFRM reaches 67.75% mAP and the highest AP_{75} in the table, 76.64%. The complete model obtains the highest mAP and AP_{50}, at 68.13% and 96.63%, while its AP_{75} is 76.61%. Relative to RF-DETR, the complete model improves the three metrics by 1.11, 1.12, and 1.48 percentage points, respectively. The results indicate complementary effects among the modules, although the best configuration varies slightly by metric.

TABLE VI
OVERALL ABLATION RESULTS OF DARQ-DET ON THE HRSC2016-HAZE TEST SET. THE FIRST ROW DENOTES THE RF-DETR BASELINE WITHOUT THE PROPOSED MODULES.

DARM	HIFRM	IQCM	Params (M)	mAP	AP_{50}	AP_{75}
–	–	–	33.36	90.46	97.53	95.50
✓	–	–	33.41	90.81	97.31	95.73
–	✓	–	33.38	90.75	97.71	95.65
–	–	✓	33.41	90.66	97.62	95.59
✓	✓	–	33.67	91.21	97.95	96.15
–	✓	✓	33.71	91.18	97.90	96.13
✓	–	✓	33.88	91.13	97.86	96.02
✓	✓	✓	34.44	91.62	98.22	96.43

HRSC2016-Haze. As shown in Table VI, each module improves mAP individually: DARM reaches 90.81%, HIFRM 90.75%, and IQCM 90.66%, compared with 90.46% for RF-DETR. The individual changes in AP_{50} and AP_{75} are not uniformly positive; for example, DARM slightly reduces AP_{50} while increasing mAP and AP_{75}. The ablation results should therefore be interpreted across all three metrics rather than through a single threshold.

The two-module combinations provide larger gains than the corresponding single-module settings. Among them, DARM+HIFRM obtains 91.21% mAP, 97.95% AP_{50}, and 96.15% AP_{75}. This result is consistent with complementarity between input-level compensation and localization-focused regression.

The complete model achieves the highest values for all three metrics: 91.62% mAP, 98.22% AP_{50}, and 96.43% AP_{75}. These values exceed the RF-DETR baseline by 1.16, 0.69, and 0.93 percentage points, respectively, indicating that the combined modules improve both average detection performance and stricter box localization on HRSC2016-Haze.

TABLE VII
OVERALL ABLATION RESULTS OF DARQ-DET ON THE RTTS TEST SET. THE FIRST ROW DENOTES THE RF-DETR BASELINE WITHOUT THE PROPOSED MODULES.

DARM	HIFRM	IQCM	Params (M)	mAP	AP_{50}	AP_{75}
–	–	–	33.36	59.90	84.60	67.60
✓	–	–	33.41	60.25	84.77	67.73
–	✓	–	33.38	60.12	84.71	67.71
–	–	✓	33.41	60.08	84.69	67.65
✓	✓	–	33.67	60.49	84.91	67.83
–	✓	✓	33.71	60.44	84.90	67.78
✓	–	✓	33.88	60.37	84.88	67.80
✓	✓	✓	34.44	60.75	85.06	68.10

RTTS. Table VII shows that DARM, HIFRM, and IQCM individually increase mAP from 59.90% to 60.25%, 60.12%, and 60.08%, respectively. The gains are small but consistent across the three individual variants, suggesting that each component contributes under real haze.

The two-module combinations further increase mAP, with DARM+HIFRM reaching 60.49%, 84.91% AP_{50}, and 67.83% AP_{75}. This pattern is consistent with complementary effects between input compensation and localization refinement.

The complete model produces the highest values in the ablation table: 60.75% mAP, 85.06% AP_{50}, and 68.10% AP_{75}. Compared with RF-DETR, the improvements are 0.85, 0.46, and 0.50 percentage points, respectively. The consistent gains across the three metrics support the use of the three modules together on RTTS.

V. CONCLUSION

This paper presents DARQ-Det, an end-to-end detector that combines degraded-input compensation, localization-focused regression, and quality-aware score calibration. DARM applies residual restoration with total variation regularization, HIFRM reweights the GIoU regression objective according to localization difficulty, and IQCM estimates box quality for candidate ranking.

On RSOD-Haze, HRSC2016-Haze, and RTTS, DARQ-Det improves mAP over the RF-DETR baseline by 1.11, 1.16, and 0.85 percentage points, respectively. The ablation results show that each module contributes to mAP and that their combination generally provides the strongest overall performance. The gains on the real-world RTTS dataset are smaller than those on the synthetic datasets but remain consistent across mAP, AP_{50}, and AP_{75}.

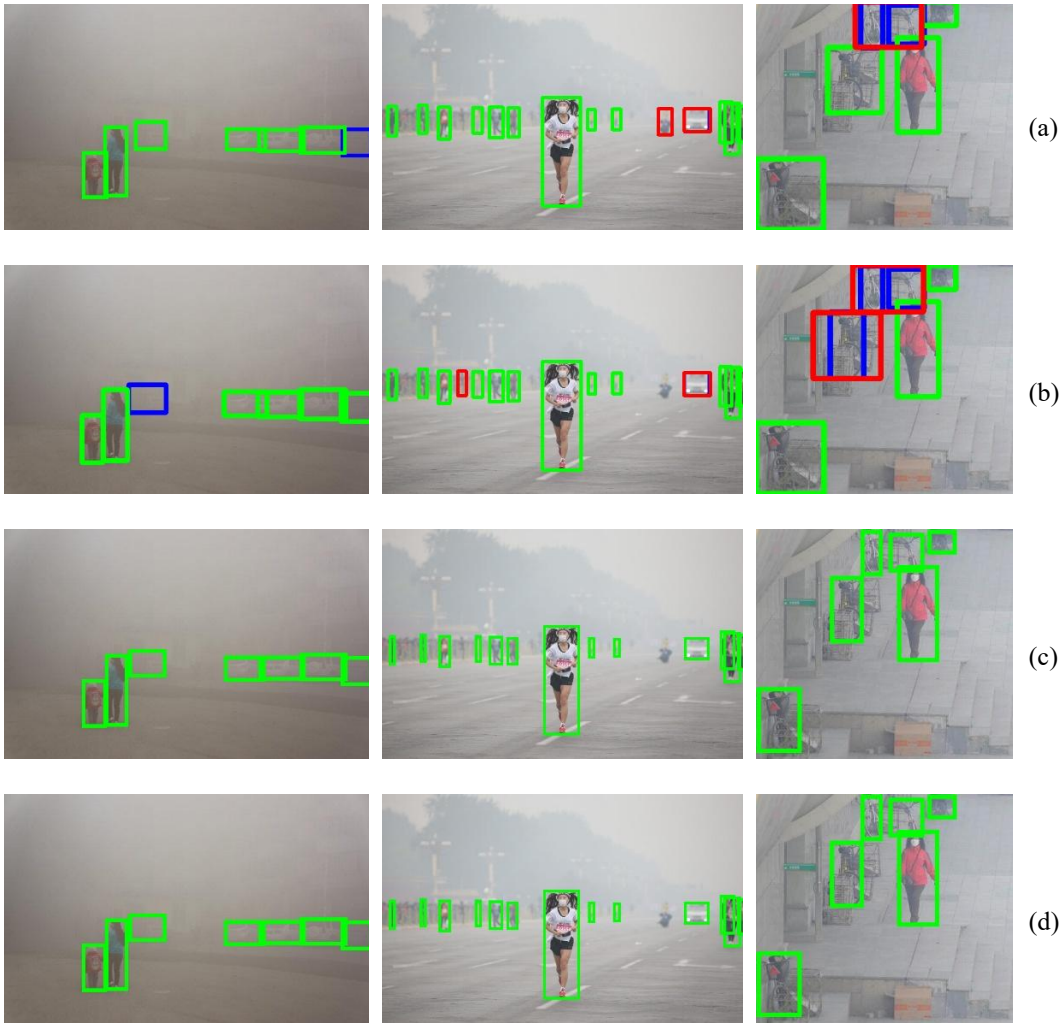


Fig. 6. Qualitative results on representative RTTS test images: (a) RT-DETR, (b) RF-DETR, (c) DARQ-Det, and (d) ground truth.

These findings indicate that joint restoration, localization refinement, and quality calibration can improve RF-DETR under haze degradation. Future work will examine a broader range of real-world degradation conditions and reduce the computational cost for resource-constrained deployment.

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